BIS x Sigma Squad AI Hackathon

BIS Standards Recommendation Engine

Fast, offline discovery of building-material standards from SP 21.

Input

product description

Output

top 3-5 BIS standards

Submission package: inference.py, retriever, local demo, public evaluation, presentation PDF

PROBLEM STATEMENT

MSEs lose time mapping products to applicable BIS standards

The current discovery task is manual: read dense SP 21 summaries, identify the right product family, then select the exact IS code and part.

For small teams, that turns a compliance question into days of document search and interpretation.

Describe	33 grade OPC cement
Search	SP 21 summaries
Recommend	IS 269: 1989

SOLUTION OVERVIEW

A judge-ready retriever that only returns standards parsed from the official PDF

Parse

Extract 574 standard summaries, titles, scopes, page references, and canonical IS IDs.

Index

Build a compact JSON index from standard IDs, title fields, scope text, and summary text.

Retrieve

Score every query with BM25 plus deterministic field and phrase boosts.

Return

Write the required JSON with top 5 standards and measured latency.

Design choice: deterministic retrieval keeps the scoring path grounded in parsed SP 21 IDs.

SYSTEM ARCHITECTURE

One offline pipeline powers both judge execution and the demo UI

dataset.pdf
BIS SP 21 source

→

build_index.py
PDF parser

→

standards.json
574 records

→

retriever.py
hybrid scorer

→

inference.py
judge output

Required judge command

```
python inference.py --input hidden_private_dataset.json --output team_results.json
```

CHUNKING & RETRIEVAL STRATEGY

Use standards as first-class records, not arbitrary page fragments

Record-level chunks

Each recommendation maps to one canonical standard ID, title, scope, summary, and source pages.

BM25 lexical core

Good fit for exact product names, material families, standard numbers, grades, and part numbers.

Field-aware boosts

Extra weight for title coverage, exact standard IDs, grade phrases, and material-specific terms.

No hallucination path

The output list is selected from parsed SP 21 IDs only; the system cannot invent an IS code.

DEMO HIGHLIGHTS

The same retriever serves CLI judging, API responses, and a local UI

White Portland cement for architectural and decorative purposes

1	IS 8042: 1989	White Portland cement
2	IS 8041: 1990	Rapid hardening Portland cement
3	IS 269: 1989	33 grade ordinary Portland cement

EVALUATION RESULTS

Public set results exceed all automated targets

100%
Hit Rate @3
target >80%

1.000
MRR @5
target >0.7

0.05s
Avg latency
target <5s

Measured by running the organizers' eval_script.py against data/public/public_test_set.json. Hidden set performance depends on the private query distribution, but the scoring path is deterministic and offline.

IMPACT ON MSES

Compliance discovery becomes guided search in seconds

Speed

Immediate standard shortlist for product planning and documentation.

Trust

Recommendations are traceable to official SP 21 summaries and page references.

Reproducibility

Offline repo runs on standard hardware with one judge command.

Team: BIS Standards Recommendation Engine Team | Acknowledgement: BIS SP 21 dataset and organizer-provided evaluation assets